

# ALLEN TU

Ph.D. Student, Computer Science — University of Maryland

Research Areas: Reliable and Scalable Vision Systems, 3D/4D Reconstruction, Generative Models

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Google Scholar: [Allen Tu](#)

## SELECTED PUBLICATIONS

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- **Allen Tu\***, H. Ying\*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, ‘[SpeedDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping](#)’. *CVPR*, 2026.
- P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, ‘[SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting](#)’. *CVPR*, 2026.
- **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, ‘[TransFIRA: Transfer Learning for Face Image Recognizability Assessment](#)’. *FG*, 2026 (Oral Presentation; Invited to submit an extended version to the IEEE T-BIOM Special Issue on the Best-Ranked Articles of IEEE FG 2026).
- A. Hanson\*, **Allen Tu\***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, ‘[PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting](#)’. *CVPR*, 2025.
- A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, ‘[Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives](#)’. *CVPR*, 2025.

\* denotes equal contribution.

## EDUCATION

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**University of Maryland, College Park**

*Ph.D. Student, Computer Science*

College Park, MD

*January 2025 – May 2028 (Expected)*

- **Advisors:** Professor Tom Goldstein, Professor Matthias Zwicker

- **Collaborators:** Professor Vishal M. Patel, Professor Rama Chellappa

*M.S. in Computer Science*

*August 2023 – December 2024*

*B.S. in Computer Science, Minor in Statistics*

*August 2019 – December 2022*

## RESEARCH EXPERIENCE

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**Amazon**

*Applied Science Intern*

Bellevue, WA

*June 2026 – Present*

- Researching methods for constructing metric-scale, photorealistic simulation environments for training embodied AI in Amazon’s Last Mile Delivery and Automation organization
- Creating scalable production pipelines for reconstructing PhysX-ready 3D digital twins from real-world sensor data passively collected by autonomous delivery platforms
- Investigating feed-forward reconstruction models and diffusion-based techniques to bridge Real2Sim and Sim2Real domain gaps across 3D modeling and perception for autonomous systems

**University of Maryland Institute of Advanced Computer Studies**

*Graduate Research Assistant*

College Park, MD

*August 2023 – Present*

- Developing scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude ([WRIVA](#)) program
- Created pruning, rasterization, and motion distillation methods that accelerate static and dynamic 3D Gaussian Splatting, producing representations with over  $10\times$  fewer primitives while preserving visual fidelity [2, 5, 6]
- Introduced uncertainty-aware diffusion and super-resolution priors into 3D reconstruction pipelines to improve fidelity under sparse and low-resolution supervision while mitigating artifacts and cross-view inconsistencies [3]

## Systems & Technology Research

Computer Vision Research Intern

Arlington, VA

May 2025 – January 2026

- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR) program
- Proposed an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [4]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at 1e-3 FMR using an encoder trained solely on public data [4]

Computer Vision Research Intern

May 2024 – August 2024

- Achieved 0.86 correlation with ground-truth probe-to-gallery similarity and state-of-the-art ERC performance for face image quality estimation (FIQA), outperforming prior methods on image-level evaluation [4]
- Designed a cluster-and-aggregate network that filters low-importance frame embeddings and fuses the remaining informative ones into a single video template, improving face recognition accuracy by up to 12%

Computer Vision Research Co-op

January 2023 – August 2023

- Implemented a mixed-voting ensemble that improved open-set search performance by 24% for face recognition, 39% for body recognition, and 49% for fused multimodal biometric recognition
- Trained a self-supervised Barlow Twins model robust to challenges like extreme distance variation and atmospheric interference, improving face recognition accuracy by 52% in severe operating conditions

Computer Vision Research Intern

June 2022 – August 2022

- Integrated pose estimation, neural novel-view synthesis, semantic segmentation, and generative inpainting to synthesize realistic cross-subject garment transfers with strong biometric identity preservation
- Augmented real training data with garment-transfer images to overcome limited garment diversity and train clothing-invariant whole-body recognition models, improving robustness to cross-garment variation

## University of Maryland Department of Computer Science

Undergraduate Researcher

College Park, MD

January 2021 – December 2022

- Analyzed 2.3 million dataset-visualization pairs to identify bias in machine learning visualization recommenders, linking model behavior to user-driven design patterns and proposing data-centric mitigation strategies [7]

## RESEARCH PROJECTS

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### IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA)

University of Maryland Institute of Advanced Computer Studies

August 2023 – Present

Unconstrained 3D reconstruction and novel view synthesis in challenging environments.

- Efficient rendering, compression, and training methods for 3D and 4D Gaussian Splatting (3DGS/4DGS) [2, 5, 6]
- Multi-view consistent super-resolution for training 3DGS with low-resolution imagery [3]
- Image, video, and multi-view diffusion model priors for sparse-view 3D reconstruction
- Uncertainty quantification algorithms for 3DGS and Neural Radiance Fields (NeRF) [5]

### IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR)

Systems & Technology Research

June 2022 – January 2026

Multimodal, opportunistic fusion of face, body, and gait information in severe operational conditions.

- Transfer Learning for Face Image Recognizability Assessment [4]
- Learned Frame Feature Aggregation for Face Recognition with Low-Quality Video
- Operating Condition-Invariant Barlow Twins and Multimodal Ensembling
- Style-Based Clothing Transfer for Robust Body Representation Learning

## HONORS & AWARDS

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**FG 2026 Doctoral Consortium:** *Selective Computation for Reliable and Scalable Vision Systems*

**University of Maryland Travel Awards:** Jacob K. Goldhaber Travel Grant, CMSC PhD Travel Grant, ICSSA

## SERVICE

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**SPAR-3D: Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models**

*Workshop Co-organizer*

*CVPR 2026*

- Co-organized the first CVPR workshop on security, privacy, and robustness in 3D generative vision models, coordinating peer review, securing \$1K in industry sponsorship, and moderating keynote and Q&A sessions

**Reviewer:** IEEE TVCG, SIGGRAPH Asia, IEEE TBIOM, CVPR Workshops, FG, IEEE TCSVT

**Conference Service:** Presented 5 accepted papers (1 oral, 4 posters) on behalf of authors unable to attend FG 2026

**Organizations:** IEEE, ACM, Computer Vision Foundation, IEEE Biometrics Council

## TEACHING EXPERIENCE

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**Peer Research Mentor**

College Park, MD

*FIRE: Capital One Machine Learning*

*January 2021 – December 2022*

- Mentored 80 undergraduate students across two cohorts in a selective multi-semester machine learning research program, guiding students from foundational training to independent research project execution
- Designed and delivered a project-driven curriculum spanning data processing, model development, training, evaluation, and deployment using modern deep learning frameworks and recent research advances
- Advised student teams end-to-end on collaborative research projects, leading to presentations in areas including text-to-audio generation, image super-resolution, and face attribute recognition

## TECHNICAL SKILLS

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**Languages:** Python, CUDA, C/C++, C#, Java, SQL, Shell

**Frameworks & Libraries:** PyTorch, TensorFlow, NumPy, SciPy, OpenCV, scikit-learn, COLMAP, Open3D, PhysX

**Specializations:** 3D/4D Reconstruction, 3D Gaussian Splatting, Neural Radiance Fields (NeRF), Differentiable Rendering, Diffusion Models, Multimodal Biometrics, Uncertainty Quantification, Vision Transformers, GPU Programming

For the full and up-to-date publication list, see [Google Scholar](#).

1. Haiyang Ying, **Allen Tu**, Jiaye Wu, Tom Goldstein, and Matthias Zwicker, ‘UniBRep: Learning Unified Geometry and Topology for Image-conditioned B-Rep Generation’. *Under review*, 2026.
2. **Allen Tu**<sup>\*</sup>, Haiyang Ying<sup>\*</sup>, Alex Hanson, Yonghan Lee, Tom Goldstein, and Matthias Zwicker, ‘[Speede3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping](#)’. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 26083–26093.
3. Pranav Asthana, Alex Hanson, **Allen Tu**, Tom Goldstein, Matthias Zwicker, and Amitabh Varshney, ‘[SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting](#)’. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 11840–11849.
4. **Allen Tu**, Kartik Narayan, Joshua Gleason, Jennifer Xu, Matthew Meyn, and Vishal M. Patel, ‘[TransFIRA: Transfer Learning for Face Image Recognizability Assessment](#)’. *Proceedings of the IEEE 20th International Conference on Automatic Face and Gesture Recognition*, 2026, pp. 1–9, doi: 10.1109/FG67764.2026.11556959 (Oral Presentation; Invited to submit an extended version to the IEEE T-BIOM Special Issue on the Best-Ranked Articles of IEEE FG 2026).
5. Alex Hanson<sup>\*</sup>, **Allen Tu**<sup>\*</sup>, Vasu Singla, Mayuka Jayawardhana, Matthias Zwicker, and Tom Goldstein, ‘[PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting](#)’. *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 5949–5958, doi: 10.1109/CVPR52734.2025.00558.
6. Alex Hanson, **Allen Tu**, Geng Lin, Vasu Singla, Matthias Zwicker, and Tom Goldstein, ‘[Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives](#)’. *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 21537–21546, doi: 10.1109/CVPR52734.2025.02006.
7. **Allen Tu**, Priyanka Mehta, Alexander Wu, Nandhini Krishnan, and Amar Mujumdar, ‘[Addressing Bias in Visualization Recommenders by Identifying Trends in Training Data: Improving VizML Through a Statistical Analysis of the Plotly Community Feed](#)’. *arXiv preprint arXiv:2203.04937*, 2022.

<sup>\*</sup> denotes equal contribution.